

ZeroDrift Mk3 — what to buy, and from where

Total ₹6,325 · ✓ = price verified against a live listing, ~ = estimate · generated from docs/data/bom_tier_a.json

ORDER THE UNDERLINED ITEMS TONIGHT. I could not see Amar Robotics' actual stock — their Instagram and IndiaMart pages are blocked from where I work — so this is routed by component *type*. The underlined five are specialist parts that a general electronics shop and a 10-minute grocery app both almost never carry, and **three of them stop the build dead if they are missing**. They are also the only items with real delivery lead time. The shop trip can happen any day; the order cannot be compressed.

1 · ORDER ONLINE NOW — ₹2,377		
Amazon.in / Robu.in / ThinkRobotics. Not a local-shop item. If the shop <i>does</i> have the AS5600 or TCA9548A, buy them there and cancel that line — but ask whether the AS5600 includes a magnet, and whether it is diametrically magnetised. Most ship bare or with the wrong type, and an axial magnet looks identical and does not work.		
2	<u>AS5600 magnetic encoder module</u>	✓ ₹498
2	<u>Diametric magnet, 6 × 2.5 mm</u>	~ ₹180
1	<u>TCA9548A I²C multiplexer</u>	✓ ₹199
1	<u>USB camera module, M12 mount, 12 mm lens</u>	~ ₹1,400
1	<u>Red filter — gel sheet or red acrylic offcut</u>	~ ₹100
subtotal		₹2,377

2 · AMAR ROBOTICS / LOCAL SHOP — ₹3,819		
Phone first for the four motion parts at the top — stocked by some shops, not all. Everything below them is standard hobby stock. The 2×AA holder replaces a coin cell; see note.		
2	NEMA17 stepper motor, 1.8°/step, 4.2 kg-cm	✓ ₹1,498
2	A4988 stepper driver module	✓ ₹340
2	NEMA17 L-bracket (pan + tilt)	✓ ₹254
2	100 µF 25 V electrolytic capacitor	~ ₹20
1	ABS project box, head enclosure	~ ₹115
1	Arduino Nano — beacon	~ ₹225
1	12 V 2 A DC power adapter	✓ ₹279
1	Acrylic sheet 3 mm, 6"×6" (pack of 2)	✓ ₹199
1	Vibration motor, 3 V DC	✓ ₹20
1	Transistor + diode + resistor combo kit	✓ ₹105
1	High-brightness RED LED, 10 mm, 650 nm	~ ₹22
1	2N2222 transistor + 220 Ω + 1 kΩ	~ ₹20
1	SPST toggle switch	~ ₹25
1	ABS project box	~ ₹115
1	High-brightness WHITE LED (brighter than beacon)	~ ₹22
1	150 Ω resistor	~ ₹5
1	2×AA battery holder + 2 AA cells — decoy supply	~ ₹70
1	Switch + scrap housing	~ ₹35
1	Dupont jumper wires — M-M, M-F, F-F	~ ₹150
1	M3 screws, nuts, standoffs assortment	~ ₹150
1	Heat-shrink, solder, hot-glue sticks	~ ₹150
subtotal		₹3,819

3 · BLINKIT / ANY KIRANA — ₹129		
Not electronics. A sports shop sells these for about a quarter of the quick-commerce price — only use Blinkit if the ten minutes matter more than ₹100.		
1	Ping-pong ball (diffuser)	✓ ₹129
subtotal		₹129

4 · ALREADY OWNED — ₹0		
No 9V battery and no coin cell: see the power note.		
—	KY-008 laser module	— ₹0
—	Arduino Nano — tracker	— ₹0
—	USB power bank (any, 5V) — beacon supply	— ₹0

- WHY NO 9 V BATTERY, AND NO COIN CELL**
- The tracker Nano runs off the laptop's USB and the motors off the 12 V adapter — neither needs a battery. The **beacon and decoy do**: the demo is two independent lights in one camera view, and a cable back to the laptop would tether them to the tracker. But a 9 V alkaline is the wrong way to do it — the Nano burns 9→5 V as heat and the cell holds ~500 mAh, so a **USB power bank you already own** is better and free. And a CR2032 is rated 0.2 mA continuous while a high-brightness LED draws ~20 mA — a hundred times over spec, so it sags and dims within about two hours. **2×AA** runs the same LED for ~100 h.
- CHECK AT THE COUNTER**
- Magnet must be DIAMETRIC**, not axial. Bring a second small magnet: a diametric one attracts on its curved *side* edge, an axial one on its flat faces.
 - Look at the motor's rear**. A short shaft out the back = dual-shaft, encoder mounts cleanly. Flat back is fine, but the sensor then needs a short, stiff overhanging arm.
 - Capacitors 25 V or higher**. 16 V on a 12 V rail has no margin against the inductive kick they exist to absorb — and skipping them kills both drivers on first power-up.
 - Camera must allow manual exposure** (UVC). Auto-exposure hunts on every blink, and the blink is the signal. Ask for 12 mm; 8 mm acceptable, 6 mm at a push.